National University of Computer and Emerging Sciences, Lahore Campus

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| Course Name: | Data Warehousing and Business Intelligence | Course Code: | DS3003 |
| Degree Program: | BS (Data Science) | Semester: | Fall 2022 |
| Exam Duration: | 60 Minutes | Total Marks: | 30 |
| Paper Date: | Wed 28 Sep 2022 | Weight | 15% |
| Section: | BDS-5A | Page(s): | 6 |
| Exam Type: | Midterm-1 | Total Questions: | 6 |

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Instructions: Scratch sheet can be used for rough work however, all the questions and steps are to be shown on question paper. *No extra/rough sheets should be submitted with question paper.* You will not get any credit if you do not show proper working, reasoning and steps as asked in question statements. *You may use a calculator.*

Q1. *(8 points)*

a. What are the essential differences between the MOLAP and ROLAP models? What are different OLAP analytical operations?

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MOLAP is basically store data in multidimensional format whereas ROLAP store data in relational format. MOLAP has a certain limit of data so it can not hold large dataset whereas ROLAP has flexibility structure which can store large data. MOLAP is performance wise good as there is no need to convert data in MD format whereas ROLAP is not good in performance wise as compare to MOLAP because ROLAP stores data in relational format so there is need to ~~[illegible]~~ convert into MD format as we have to represent data in cube. 2

OLAP Analytical Operations: Drill through, Rollup, Drill Down, Drill Across

b. What are virtual cubes? Why are they needed?

When the cube is so large beyond the capacity (max) so we make virtual cubes e.g we need data from 2 cubes but if we combined them so it will exceed the limit. We make virtual cube at runtime from different cubes although join operation is costly as it takes more time but we get answer of our business question. 2

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c. What are the various data sources for the data warehouse?

Product : data from current OLTP (online data)
Internal : private data (e.g diary, notepad)
external : data from external agency
Archived : offline data ( data of previous years) , firstly we will online it then used for our data warehouse.

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d. What is the concept of factless fact table? Give an example.

Factless fact table is a fact table which contain PK of dimension tables as FK in fact table but there is no pre aggregate fact in fact table.

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e.g

| |
|---|
| Product ID |
| Store ID |
| Time ID |

There is no facts in this table

. (3 points) Consider the following normalized data structure: **Roll No:** ____________

SALES (saleId, storeId, saleDate, ...)

SALES_DETAIL (<u>transactionId</u>, saleId, itemId, itemQty, ...)

Assume there are one million sales and ten million sales details. Record length of both tables is same i.e., 100 bytes and each column of both tables including PK/FK column is of same size i.e., 10 bytes.

Query: SELECT * FROM sales S JOIN sales_detail D ON S.saleId = D.saleId;

You are required to improve the performance of the above query using pre-join de-normalization technique. Show your de-normalized data structure schema and evaluate increase in additional storage cost (in megabytes or %age) for the de-normalized data structure.

Sales → $10^6$ records

Sales detail → $10 \times 10^6$ records

Storage cost: $(10^6 \times 100) + (10 \times 10^6 \times 100)$

$= (10^8 + (10^9))$ Bytes $= 1100$ MB

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Pre-join denormalization:

There will be $10 \times 10^6$ records

Sales has: $\frac{100}{10} = 10$ columns

Salesdetail has: $\frac{100}{10} = 10$ columns

Now there will be 19 columns

Storage cost: $10 \times 10^6 \times 19 \times 10$

$= 19 \times 10^8$ Bytes

$= 1900$ MB

Increase in additional storage cost: $1900 - 1100 = 800$ MB

Roll No:

Consider the following case study for the next two questions:

**Customer Dim**: Customer Code, Customer Desc, City ID, City Desc, Country ID, Country Desc
**Sales Rep Dim**: Sales Rep No, Sales Rep Desc, Organization ID, Organization Desc, Channel ID, Channel Desc
**Bill Date Dim**: Bill Date, Bill Day Desc, Bill Month ID, Bill Month Desc, Bill Year ID, Bill Year Desc
**Rate Plan Dim:** Rate Plan Id, Rate Plan Desc, Rate Plan Type Code, Rate Plan Type Desc
**Billing Fact**: Bill Date, Customer Code, Sales Rep No, Rate Plan Id, No of Calls, No of Total Minutes, Taxes, Regulatory Charge

**Assume:** 100,000 customers, 60 cities, 3 countries, 100 sales rep, 4 organizations, 2 channels, 30 rate plans, 3 rate plan types, and 5 years billing history.

**Q3.** *(2 points)* Identify the full-additive, semi-additive, and non-additive facts, if any, in the above billing base fact table.

full additive : No of calls, No of Total Minutes, Taxes, Regulatory Charge

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Semi- additive : ~~No of calls~~

non-additive :

**Q4.** *(3 points)* Refer to the above Sales Rep dimension schema. Show the revised Sales Rep dimension schema that also preserves the history of changes to the Sales Rep.

To preserve the history ~~use Type 2~~ :

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Sales Rep Dim : Sales ID, Sales Rep No, Sales Rep Desc, Organization ID, Organization Desc, Channel ID, Channel Dec, effective date, ending date

Sales ID will be a surrogate Key/ artificial key & we have added starting / effective date & ending date column.

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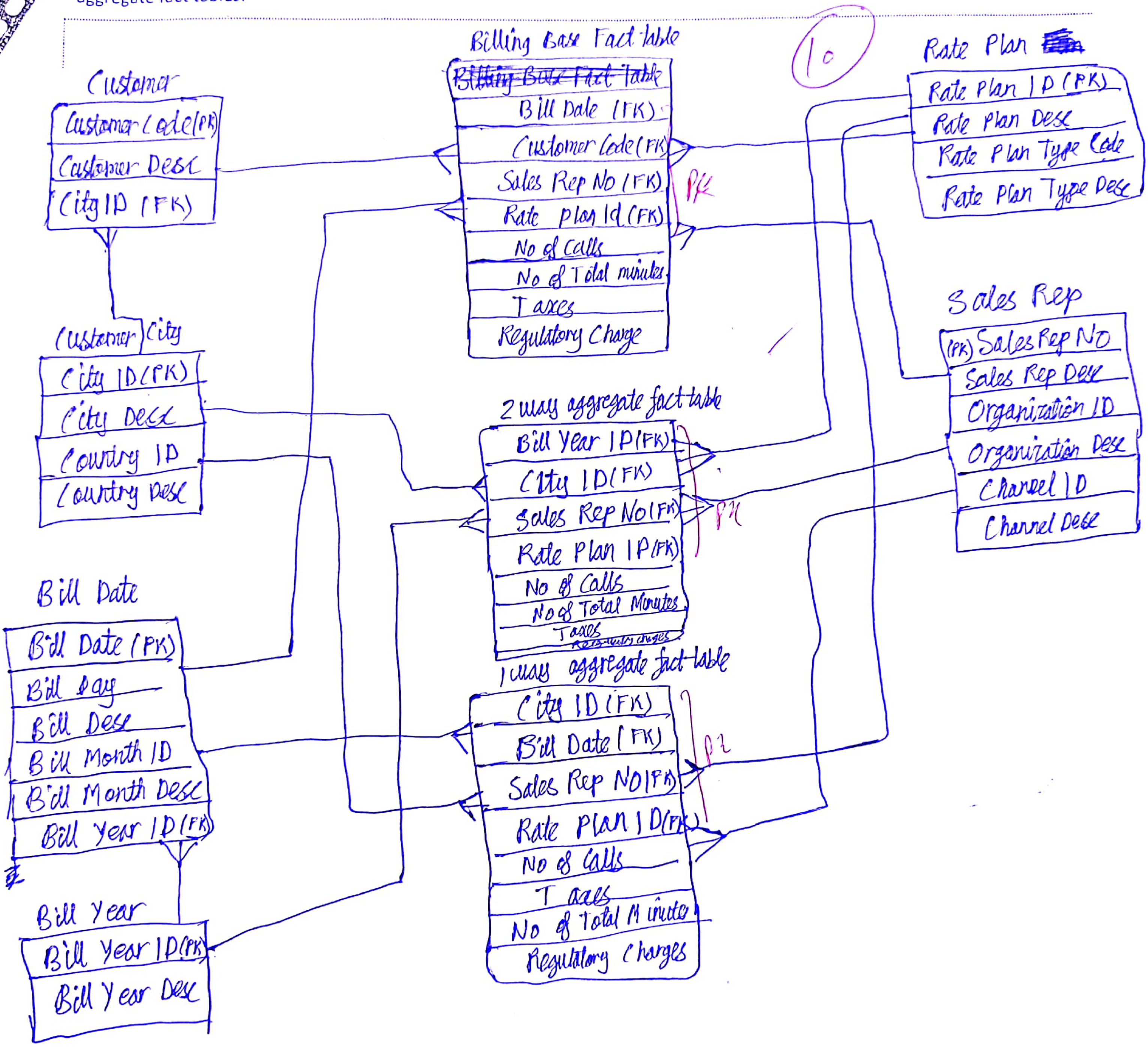
15. *(10 points)* Draw the appropriate star schema that includes a base fact table, a 1-way aggregate fact table (along Customer City), and a 2-way aggregate fact table (along Bill Year and Customer City). Show the primary keys, foreign keys and all the relationships between the dimensions and fact tables. Note: Draw only one diagram that includes base fact table as well as aggregate fact tables.



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Q6. *(4 points)* Estimate the size (in number of rows) of the above sales rep dimension table, billing base fact table, and both aggregate fact tables.

Number of rows:

Sales rep dim : 100

Billing Base fact table : $5.475 \times 10^{11}$

1 way aggregate fact table : $3.285 \times 10^{11}$

2 way aggregate fact table : $9 \times 10^{5}$

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